

Mostafa Hamdy

Cairo-Egypt | (+20) 1099340306 | mostafaboshta9@gmail.com | [in/Mostafa-Hamdy](https://www.linkedin.com/in/Mostafa-Hamdy) (LinkedIn)

OBJECTIVE

Highly motivated Data Engineer with hands-on experience in data processing, pipeline development, and analytics. Eager to further develop my career by contributing to a dynamic environment where I can leverage my expertise in data engineering tools and techniques.

EDUCATION

Data Engineering Track - Postgraduate degree

(May 2024 - Sep 2024)

Information Technology Institute (ITI), Ministry of Communications and Information (MCIT)

Bachelor of Computer Science and Engineering Grade

(Aug 2017 - July 2022)

Menoufia University

Grade: Good

Graduation Project: Driver Drowsiness Monitoring System (Excellent)

Experience

Software Engineer @ Wataniya for Roads. (Military Service)

(March 2023 - March 2024)

- Gathered system requirements from stakeholders.
- Created a plan for the software's architecture and design.
- Developed the software application and wrote code.
- Executed test cases to identify and fix bugs or errors.
- Released software meeting high quality standards.

COURSES & CERTIFICATES

- Associate Data Engineer in SQL Track (from DataCamp)
- Gold badge in SQL (from HackerRank)
- Gold badge in Python (from HackerRank)
- SQL fundamentals Track (from DataCamp)
- Implementing Analytics Solutions Using Microsoft Fabric (from Microsoft)
- Machine Learning Specialization (from Coursera)

TECHNICAL SKILLS

- | | | | |
|--------------------|------------------|-----------|-------------------------|
| • Database Design | • RDBMS | • SQL | • Python |
| • Data warehousing | • Data Modeling | • ETL | • Git & version control |
| • Hadoop | • Pyspark | • Docker | • Cloud computing |
| • Kafka | • Apache Airflow | • Jenkins | • Web scraping |

PROJECTS

Movie Platforms [\[GitHub\]](#)

This project extracts data from various streaming platforms (Netflix, Hulu, Amazon Prime, Disney+) and sends it to Kafka for processing using Apache Spark. The data producers are managed and run in parallel using Apache Airflow. The processed data is then streamed into a database table and visualized on a real-time streaming dashboard.

Technologies used: **Apache Airflow, Apache Kafka, Apache Spark, SQL Server, Power Bi, Git.**

Gravity Books Data [\[GitHub\]](#)

The Gravity Books ETL Project utilizes **SQL Server Integration Services (SSIS)** to perform data extraction, transformation, and loading (ETL) of book sales and order tracking systems from an OLTP source into a centralized Data warehouse. This project is designed to streamline data flow and support analytics for book inventory, sales, and customer transactions.

Technologies used: **SQL Server, SSIS, Git.**

Bank System Datawarehouse [\[GitHub\]](#)

This project demonstrates the use of SQL Server Integration Services (SSIS) for building an ETL (Extract, Transform, Load) pipeline for a banking system, alongside using Power BI for reporting and visualizations.

Technologies used: **SQL Server, SSIS, Power BI, Git.**

Driver Drowsiness Monitoring System [\[GitHub\]](#)

This project is focused on developing different machine learning and deep learning models to detect driver drowsiness using data-driven methods. The goal is to improve road safety by identifying signs of fatigue and alerting drivers before dangerous situations arise.

Technologies used: **Python, Traditional Machine Learning (SVM), OpenCV, Transfer Learning**

Global Terrorism EDA [\[GitHub\]](#)

Global Terrorism EDA is a data analysis project that aims to explore and visualize terrorism incidents worldwide. The goal is to provide insights into the nature, frequency, and characteristics of terrorist activities across different regions and time periods.

Technologies used: **Python, Pandas, Matplotlib, Plotly, Dash**

Amazon Product Sentiment Analysis [\[GitHub\]](#)

This data science project focuses on analyzing and extracting insights from Amazon product reviews to provide valuable feedback. The goal is to develop a system that can automatically categorize and analyze customer reviews.

Technologies used: **Python, Pandas, Matplotlib, Plotly, Scikit-learn, Numpy**

Achievements

- Best Engineering Idea in FEE Control Exhibition Competition. (June 2022)
- 1st Place in NASA Space Apps Global Competition. (June 2021)
- Global Nominee in Nasa Space Apps Global Competition. (June 2021)